

Online Agriculture Products Store

Q1. 4 Quarterly Audits are planned Q1 , Q2, Q3, Q4 for this Project. What is your knowledge on how these Audits will happen for a BA ?

Stage	Requirement Gathering phase-15 weeks (Week 1 to Week 15)
Completed	10 weeks (Week1 to Week 10)
Checklist	BRD template
	Elicitation results report
	Duplicate requirements/features - client signoff
	Email communication - To, CC, BCC

Stage	Requirements Analysis phase-13 weeks (Week 16 to Week 29)
Completed	7 weeks (Week 16 to Week 23)
Checklist	Stakeholder analysis report
	Functional and non functional documents mapping
	UML diagrams
	Client signoff on documents
	Email communication - To, CC, BCC

Stage	Design phase-10 weeks (Week 30 to Week 40)
Completed	9 weeks (Week 29 to Week 38)
Checklist	Utilization of tools on client communication
	Stakeholder MoM
	Documented evidence on client communication
	Email communication - To, CC, BCC

Stage	Development Phase-12 weeks (Week 41 to Week 52)
Completed	10 weeks (Week 41 to Week 51)
Checklist	JAD sessions report
	End user manual preparation document
	BA and developer MOM
	Email communication - To, CC, BCC

Stage	Testing Phase-8 weeks (Week 53 to Week 63)
Completed	6 weeks (Week 52 to Week 58)
Checklist	Testing case summary
	Integration test report
	System test defect log
	Training report to end users
	Email communication - To, CC, BCC

Stage	UAT & Project Closure-12 weeks (Week 40 to Week 52)
Completed	10 weeks (Week 40 to Week 50)
Checklist	UAT defect closure
	Client project acceptance form
	Final RTM completeness (100% BR coverage)
	RACI closure
	Email communication - To, CC, BCC (To: Mr. Henry/Committee, CC: All stakeholders, BCC: APT IT Solutions (go-live MoM, handover docs)

Q2. Before the Project is going to Kick Start, The Committee asked Mr Karthik to submit BA Approach Strategy

Write BA Approach strategy (As a business analyst, what are the steps that you would need to follow to complete a project – What Elicitation Techniques to apply, how to do Stakeholder Analysis RACI/ILS, What Documents to Write, What process to follow to Sign off on the Documents, How to take Approvals from the Client, What Communication Channels to establish n implement, How to Handle Change Requests, How to update the progress of the project to the Stakeholders, How to take signoff on the UAT- Client Project Acceptance Form)

Your Team

Project Manager - Mr Vandanam

Senior Java Developer - Ms. Juhi

Java Developers - Mr Teyson, Ms Lucie, Mr Tucker, Mr Bravo

Network Admin - Mr Mike

DB Admin - Mr John

Testers - Mr Jason and Ms Alekya

BA - You

Technical Team have assembled to discuss on the Project approach and have finalised to follow 3-tier architecture for this project.

1. Elicitation Techniques to Apply

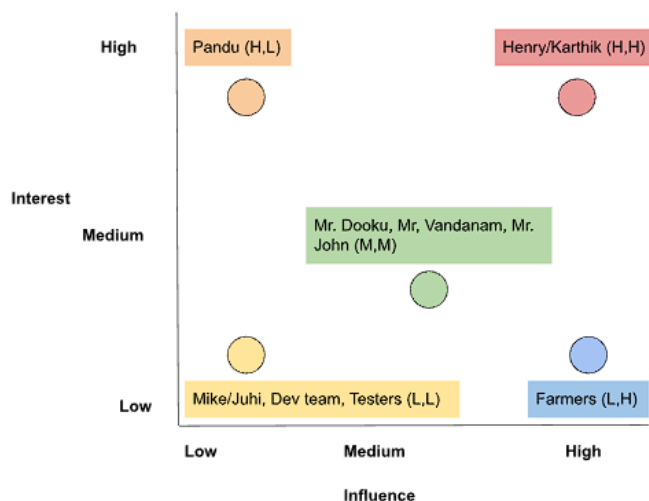
- **Primary:**
 - Interviews (Mr. Henry for vision, Peter/Kevin/Ben for farmer pain points like procurement delays)
 - JAD workshops (committee + manufacturers for catalog/payment flows)
 - Prototyping (mock login/search UI for remote farmer validation).
- **Secondary:**
 - Document analysis
 - Brainstorming
 - Surveys (Farmer survey to understand COD/UPI preferences).

Rationale: Mixed methods capture high-level business needs + detailed usability for agri users; avoids bias.

2. Stakeholder Analysis (RACI/ILS)

Stakeholder analysis identifies key players' roles, influence, and interests to prioritize engagement for the Online Agriculture Products Store project.

- RACI Matrix (Responsible, Accountable, Consulted, Informed)
- ILS (Influence/Interest)
 - Manage Closely (Top-Right): Frequent JADs, prototypes (Mr. Henry - Sponsor, Mr. Karthik - Delivery Head).
 - Keep Satisfied (Bottom-Right): Regular demos, feedback loops (Peter/Kevin/Ben - Farmers/Key Stakeholders).
 - Keep Informed (Top-Left): Quarterly audits, dashboards (Mr. Pandu - Financial Head).
 - Monitor (Bottom-Left): Monthly status emails (Mr. Mike - Network Admin, Ms. Juhi - Senior Java Developer, Mr. Teyson/Ms. Lucie/Mr. Tucker/Mr. Bravo - Java Developers, Mr. Jason/Ms. Alekya - Testers).
 - Medium/Medium (Center): Develop Understanding (Mr. Dooku-Project Coordinator, Mr. Vandanam - Project Manager, Mr. John - DB Admin): Bi-weekly dashboards, targeted JAD consults, monthly MoM reviews.



3. Documents to Write

- BRD (Business requirement document): 10+ prioritized business requirements (BR001: Farmer search, BR002: Manufacturer upload)
- SRS/FRS (System/Functional requirement specifications): Functional specs with 3-tier mappings (UI: catalog, Logic: payments, Data: inventory)
- Stakeholder Register & RACI, Assumptions log, Change Request log, UAT plan
- RTM (Requirement Traceability Matrix): Traceability matrix (BRs → use cases → test cases)
- UML (Unified Modelling Language) Artifacts: Use case diagrams/specs (5+), activity diagrams (login→order→track)

4. Process to Sign Off on Documents

- Draft Review: Circulate documents to RACI "Responsible" parties (e.g., farmers for BRD usability) via tracked email.
- Walkthrough Meeting: 1-hr JAD with Mr. Vandanam + reviewers; capture feedback in MoM.
- Revise & Redistribute: v1.0 with change log; 3-day review window.
- Formal Signoff: Digital signature page (Adobe Sign) + email confirmation from Accountable (Mr. Karthik/Henry).
- RTM Update: Link signed doc to baseline requirements

5. How to Take Approvals from Client

- Escalation Path: BA → Mr. Vandanam → Mr. Karthik → Committee (Henry/Pandu/Dooku).
- Approval Gates before proceeding with the next phase: Q1 BRD (W15), Q2 Analysis (W21), Q3 Design (W39), Q4 UAT (W69).
- Method: Formal MoM emails with attachments + MS Teams approval workflow; physical signoff for ₹2 Cr milestones.
- Fallback: 48-hr escalation if no response; conditional approval for non-critical changes

6. Communication Channels to Establish & Implement

Channel	Purpose	Frequency	Participants
MS Teams	Daily standups, quick queries	Daily 15-min	Full team
Email	Formal approvals, MoMs	As-needed	To:Approvers, CC:Team
JAD Workshops	Elicitation, reviews	Bi-weekly	Stakeholders + Core team
Jira/Confluence	RTM tracking, docs	Real-time	Devs + BA + Testers
Quarterly Audit Calls	Milestone gates	End Q1-Q4	Committee + Full team

BA maintains comms log in timesheets.

7. How to Handle Change Requests

- Log: New/updated req in Change Req Register (ID, impact on scope/budget/timeline, BR reference).
- Impact Analysis: BA assesses (e.g., new "geolocation tracking" = +2 weeks Dev, +₹5L).
- Change request Review: Present to Mr. Vandanam + Ms. Juhi; get committee approval if >5% scope.
- RTM Update: Baseline changes only post-approval; notify via email.
- Trace Forward: Update UML/test cases; no changes post-UAT freeze (W58)

8. How to Update Progress to Stakeholders

- Weekly Dashboard: Jira burndown + RTM coverage % emailed to Mr. Karthik/Mr. Vandanam.
- Bi-Weekly MoM: Post-JAD summary (achievements, risks, next steps) to full RACI.
- Quarterly Audits: Formal presentations (W15/21/39/69) with checklist signoff.
- Escalation Reports: Red/Yellow/Green status; risks >Med to committee within 24 hrs

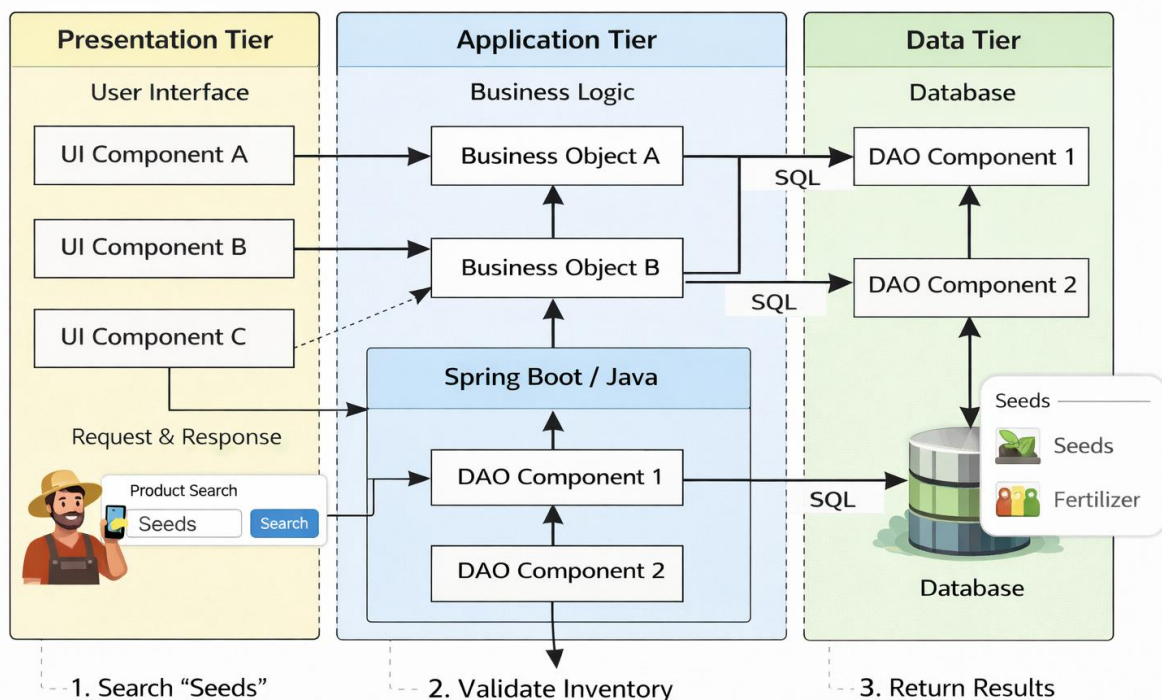
9. UAT Signoff Process (Client Project Acceptance Form)

- Preparation (W58): BA derives test cases from use cases; train farmers (Peter/Kevin/Ben) on test scenarios.
- Execution (W60-66): 95% pass rate target; daily defect log shared via Teams.
- Exit Criteria: Zero critical defects reported; 100% BR coverage validated.
- Acceptance Form: Pre-filled checklist (BR001-010 tested, performance meets Non functional requirements); digital sig from Mr. Henry.
- Formal Handover: Email To: Committee (form + RTM + support contacts), CC: All, BCC: APT legal.
- Go/No-Go: Mr. Karthik chairs gate meeting to take a final decision on deployment (W69); triggers W70 deployment

Q3. Explain and illustrate 3-tier architecture?

3-tier architecture separates an application into Presentation (UI), Application (business logic), and Data (database) layers for scalability and maintainability.

1. **Presentation Tier: Farmer/manufacturer UI** (React/HTML)
 - a. Login screens, product catalog, search "fertilizers"
 - b. Team: Front-end devs render BR001 (farmer search)
2. **Application Tier: Java business logic** (Spring Boot)
 - a. Payment processing (COD/UPI), inventory validation
 - b. Team: Ms. Juhi implements BR002 (manufacturer upload)
3. **Data Tier: Database storage** (MySQL by Mr. John)
 - Products table, farmer profiles, order history
 - Team: Mr. John manages seeds/fertilizers schema



Flow: Peter (farmer) searches seeds → Java validates stock → DB returns results → UI displays catalog → COD order confirmed.

Q4. Business Analyst should keep What points in his/her mind before he frames a Question to ask to the Stakeholder.

Business analysts should frame stakeholder questions thoughtfully to elicit clear, actionable requirements aligned with project goals and modeling needs. Key considerations draw from structured techniques like 5W1H, SMART criteria, RACI roles, and artifacts such as use cases and diagrams

5W1H Framework

Use Who, What, When, Where, Why, and How to probe comprehensively for consistent requirements. This tool ensures questions cover context, such as "Who performs this process?" or "How does the current system handle exceptions?" to avoid gaps in understanding business processes.

SMART Criteria

Ensure questions yield Specific, Measurable, Achievable, Relevant, and Time-bound responses for validating requirements. For example, frame queries to confirm if a need is feasible within constraints, like "What measurable outcome defines success for this feature by Q2?"

RACI Alignment

Tailor questions based on stakeholder roles: Responsible (doers), Accountable (owners), Consulted (experts), Informed (updates). Ask Responsible parties for execution details, Consulted SMEs for domain insights, clarifying involvement to prevent overlaps.

Modeling Artifacts

Consider how answers feed into use cases, specs, activity diagrams, models, and page designs within 3-tier architecture (presentation, logic, data layers). Questions should target flows, like "What user interactions occur in the UI layer?" to build precise UML diagrams and specs.

Technique	Purpose in Question Framing	Example Question
5W1H	Extract full context	Why does this exception trigger now?
SMART	Ensure verifiable needs	Is this achievable with current tech by deadline?

RACI	Match role to input	As the Accountable owner, what decisions do you approve?
Artifacts	Link to deliverables	How does this fit activity diagram swimlanes?

Q5. As a Business Analyst, What Elicitation Techniques you are aware of? (BDRFOWJIPQU)

Common techniques include structured approaches to uncover needs efficiently.

- Brainstorming: Group sessions to generate ideas rapidly without criticism, ideal for creative requirements.
- Document Analysis: Review existing docs like manuals or processes to identify gaps and context.
- Reverse Engineering: Extracts requirements from legacy code (black box or white box), common in migration projects.
 - Black Box reverse engineering: The product is studied without examining the internal structure
 - White Box engineering: The product is studied with the inner workings of the system.
- Requirements Workshops: Facilitated group meetings for collaborative consensus on needs.
- JAD: Joint Application Development is an extended facilitated workshops for stakeholder collaboration and rapid requirements consensus.
- Focus Groups: Interactive sessions (6-12 attendees) where moderated participants share ideas, preferences, and needs about products/services.
 - Homogeneous (similar backgrounds): Limited perspectives - use separate sessions.
 - Heterogeneous (diverse backgrounds): Self-censorship risk - lower data quality.
- Observation: Watch stakeholders at work (active or passive) to spot unstated processes.
 - Passive/Invisible: Silent observation, questions after completion (multiple sessions).
 - Active/Visible: Real-time dialog, participate in questions while ongoing shadowing.

These complement the core set for diverse scenarios.

- Interviews: One-on-one or small group talks using open/closed questions for depth.

- Prototyping: Build mockups to validate UI/flows iteratively with feedback.
- Questionnaires/Surveys: Reach many stakeholders quickly for quantitative input.
- Use Case Analysis: Use case analysis identifies actors, scenarios, and system interactions to derive functional requirements and test cases.

Acronym Fit	Technique	Best For
B	Brainstorming	Idea generation
D	Document Analysis	AS-IS processes, gap analysis
R	Reverse Engineering	Legacy code extraction (migrations)
F	Focus Groups	Attitudes, preferences (6-12 people)
O	Observation	Unclear explanations, real processes
W	Workshops	Consensus, high-quality requirements
J	JAD	Large info consolidation, discrepancies
I	Interviews	Detailed exploration, clarification
P	Prototyping	Visualization, early validation
Q	Questionnaires	Remote/large groups, quick input
U	Use Case Analysis	Functional specs, test cases

Q6. Which Elicitation Techniques can be used in this Project and Justify your selection of Elicitation Techniques?

Prototyping

Use case Specs

Document Analysis

Brainstorming

Prototyping

Create interactive mockups of the 3-tier platform (UI for farmer listings, logic for recommendations, data for inventory) to visualize features like C2C trading and SMS notifications and provide UI based mockups. Justification: Farmers may struggle articulating mobile-first needs; prototypes enable rapid feedback, reducing rework by 30-40% in agriculture e-commerce where UI drives adoption

Use Case Specs

Develop detailed specs for core scenarios (e.g., "Farmer posts product," "Retailer searches by rating/context"). Justification: Aligns with V-Model's requirements analysis phase, capturing explicit flows for Java devs/testers, ensuring traceability in fixed-budget sequential development.

Document Analysis

Review existing docs like farmer logs, market reports, govt. schemes (e.g., digital agriculture push), and prior RFPs. Justification: Uncovers baseline processes (e.g., middlemen margins, 20-30% post-harvest losses) without stakeholder time, efficient for dispersed rural users and budget limits.

Brainstorming

Facilitate sessions with Mr. Karthik, SMEs, farmers, and retailers for innovative ideas like AI recommendations or yield analytics. Justification: Generates creative solutions for unique pain points (e.g., transparent pricing), fostering buy-in in early RG/RA phases before design.

All four methods are eligible for elicitation however Prototyping stands out as the best elicitation technique among the give options for the Online Agriculture Products Store project.

Why Prototyping?

It delivers tangible mockups of the 3-tier e-commerce platform (e.g., farmer app screens for product listing, retailer dashboards), allowing non-tech stakeholders like farmers (Peter/Kevin/Ben) to interact and provide precise feedback on usability. In V-Model's early phases, it uncovers hidden needs (e.g., SMS for low-connectivity areas) faster than abstract specs or docs, cutting rework by 30-40% under ₹2 crore budget constraints.

Comparison to Others

Use case specs and document analysis provide structure but lack visuals for rural users; brainstorming sparks ideas without validation. Prototyping bridges this by

iterating quickly, aligning with sequential SDLC and prior emphasis on prototypes post-interviews/JAD.

Technique	Strength	Limitation vs. Prototyping
Prototyping	Visual validation, low rework	Best overall
Use Case Specs	Traceability	Too textual for farmers
Document Analysis	Cost-effective baseline	Misses tacit needs
Brainstorming	Creativity	No hands-on testing

Q7. Fertilizers, seeds, pesticides details from the manufacturers and should be able to display them to the Farmers.

To gather the business requirements from the client, you went to SOONY and met Mr. Henry. When Mr. Henry was asked about the project and what are they expecting from the project, Mr. Henry stated that he is expecting to have a login for all its users (fertilizers, seeds, pesticides manufacturers and Farmers) , a product catalog of fertilizers, seeds, pesticides, a search option to search for products, payment process, and delivery tracking.

After doing the stakeholder analysis, you have found out that Peter, Kevin, Ben are the key stakeholders and you have scheduled an appointment to meet them. After meeting with them and trying to gather the stakeholder requirements, Kevin said that, a Farmer should be able to browse through the products catalog once they visit the website and need to have a search option so that they can search for any product they need. Peter said that, if a farmer wants to buy any product or add them to buy-later list, they need to login first using their email id and password. If it is a new user, then they can create a new account by submitting their email ID and creating a secure password. Ben added saying that, Farmers needs to have an easy-to-use payment gateway which should include cash-on-delivery (COD), Credit/Debit card and UPI options so that the user's experience should be better. Kevin mentioned that, a user gets an email confirmation regarding their order status. A delivery tracker to track the whereabouts of their order.

Identify Business Requirements (which includes Stakeholder Requirements)
BR001 – Farmers should be able to search for available products in fertilizers,

seeds, pesticides

BR002 – Manufacturers should be able to upload and display their products in the application

Make suitable Assumptions and identify at least 10 Business Requirements.

Business requirements for the Online Agriculture Products Store have been identified from Mr. Henry's high-level expectations and key stakeholders Peter, Kevin, and Ben's inputs. These capture core needs for manufacturers and farmers across authentication, catalog, search, payments, and tracking.

The following 12 prioritized requirements emerge directly from stakeholder discussions:

- BR001 – Farmers should be able to search for available products in fertilizers, seeds, pesticides [given]
- BR002 – Manufacturers should be able to upload and display their products in the application [given]
- BR003 – All users (manufacturers, farmers) must have login functionality using email and password
- BR004 – New users must be able to create accounts by submitting email ID and secure password
- BR005 – Farmers can browse complete product catalog without requiring login
- BR006 – Farmers can add products to a "buy-later" wishlist after login
- BR007 – Payment gateway must support Cash-on-Delivery (COD), Credit/Debit cards, and UPI options
- BR008 – Users receive email confirmations for all order status updates
- BR009 – Farmers can track delivery status and order whereabouts in real-time
- BR010 – System must maintain product catalog for fertilizers, seeds, and pesticides from manufacturers
- BR011 – Search functionality available across all product categories with relevant filters
- BR012 – Application must support mobile-responsive design for rural farmers' accessibility

Q8. List your assumptions

- Farmers have basic smartphones with internet access for app usage
- Manufacturers have digital product details (images, specs) ready for upload
- Payment gateway integration uses existing providers (Razorpay/Paytm) supporting COD/UPI
- Email notifications leverage standard SMTP services; no custom SMS initially

- 3-tier architecture handles expected traffic (10K farmers initially registering or accessing the portal)
- Basic security (HTTPS, password hashing) meets regulatory needs
- V-Model timeline accommodates iterative refinement of these requirements during RA phase

Q9. Give Priority 1 to 10 numbers (1 being low priority – 10 being high priority) to these Requirements after discussions with the stakeholders.

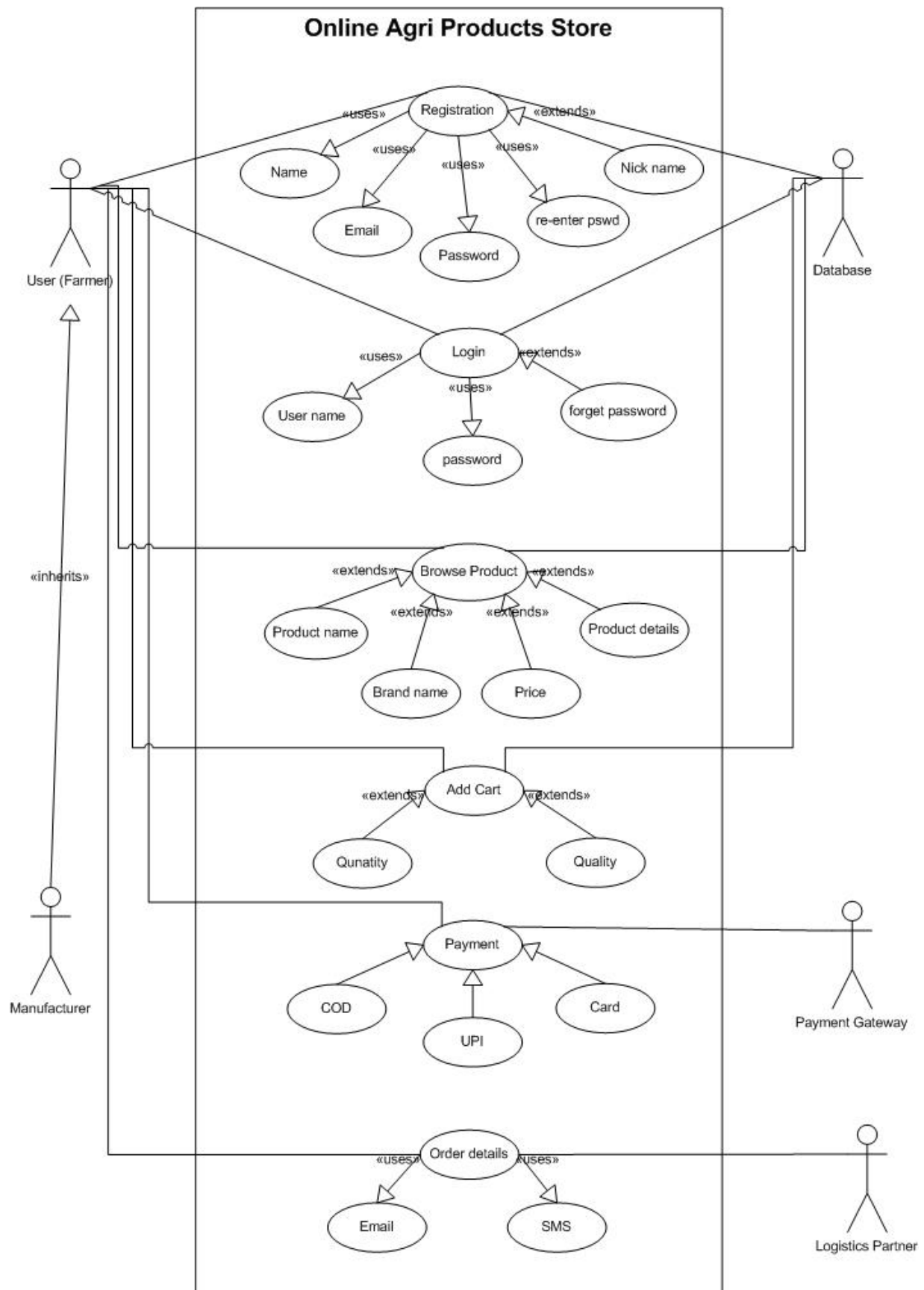
Req ID	Req Name	Req Description	Priority
BR001	Farmer Search	Farmers should be able to search for available products in fertilizers, seeds, pesticides	10
BR002	Manufacturers upload their Products	Manufacturers should be able to upload and display their products in the application	9
BR003	User Login	All users (manufacturers, farmers) must have login functionality using email and password	10
BR004	New User Registration	New users must be able to create accounts by submitting email ID and secure password	9
BR005	Product Catalog Browse	Farmers can browse complete product catalog without requiring login	8
BR006	Buy-Later Wishlist	Farmers can add products to a "buy-later" wishlist after login	7
BR007	Payment Gateway	Payment gateway must support Cash-on-Delivery (COD), Credit/Debit cards, and UPI options	10
BR008	Order Confirmation Email	Users receive email confirmations for all order status updates	8
BR009	Delivery Tracking	Farmers can track delivery status and order whereabouts in real-time	9
BR010	Product Catalog Management	System must maintain product catalog for fertilizers, seeds, and pesticides from manufacturers	9
BR011	Advanced Search Filters	Search functionality available across all product categories with relevant filters	8
BR012	Mobile Responsive Design	Application must support mobile-responsive design for rural farmers' accessibility	9

Priorities assigned post-stakeholder discussions (Mr. Henry, Peter, Kevin, Ben):

- Core commerce functions (search, login, payments) rank 10 as they're foundational to platform viability and farmer adoption.
- Manufacturer upload and tracking rank 9 for supply chain completeness within ₹2 crore budget.
- Lower priorities (7-8) for enhancements that build on MVP success.

Q10. Draw use case diagram

- Actors:
 - Primary: Farmers (Peter, Kevin, Ben)
 - Secondary: Manufacturers as primary external users, Database, Payment Gateway and Logistics Partner
- Use Cases:
 - Essential use cases: Mapped directly to business requirements with registration, Logic, Product catalog, Add Cart, Payment and Order details.
 - Supporting use case: Help essential use case complete their process
- System Boundary: Clearly separates the 3-tier platform from external stakeholders
- Relation:
 - Extend: Shows optional relationships
 - Include: Shows mandatory relationships
 - Generalization: Shows inheritance from child to parent actor attributes



Q11. Prepare use case specs for all use cases

Use Case 1: Farmer Search Products

Use Case ID	UC001		
Use Case Name	Farmer Search Products		
Priority	10 (Critical)		
Created By	Mr. ABC	Last Updated By	January 28th, 2026
Date Created	January 19th, 2026	Last revision date	January 25th, 2026
Actor	Primary - Farmer, Secondary - System		
Description	Farmers search for fertilizers, seeds, and pesticides across the product catalog.		
Pre-condition	User accesses platform (guest or logged in)		
Post-condition	Search results displayed with applied filters		
Trigger	Farmer enters search keywords or selects category		
Normal Flow of events/ Basic Flow/ Happy Path	1. Farmer navigates to search/catalog page 2. Farmer enters keywords OR selects category filter 3. System queries product database (fertilizers/seeds/pesticides) 4. System returns matching results sorted by relevance 5. Farmer views product details (price, specs, ratings)		
Alternative Flow	A1: No results found → System shows "related products" A2: Farmer applies additional filters → Results refined		
Exceptions	E1: Network error → Display cached results E2: Invalid search → Suggest corrections		
Special Requirements	Mobile responsive, <2 sec response time		
Assumptions	Basic internet connectivity available		
Frequency of Use	1000 searches/day		

Use Case 2: User Login

Use Case ID	UC004		
Use Case Name	User Login		
Created By	Mr. ABC	Last Updated By	January 28th, 2026
Date Created	January 19th, 2026	Last revision date	January 25th, 2026
Actor	Farmer/Manufacturer, Secondary - System		
Description	Authenticated users access personalized platform features.		
Pre-condition	User has valid registered account		
Post-condition	User session established, role-based dashboard loaded		
Trigger	User clicks "Login" button		
Normal Flow of events/ Basic Flow/ Happy Path	1. User navigates to login page 2. User enters registered email and password 3. System validates credentials against database 4. System creates secure session 5. System redirects to role-specific dashboard		
Alternative Flow	A1: "Remember Me" → Persistent cookie created A2: Forgot password → Reset link emailed		
Exceptions	E1: Invalid credentials (3 attempts) → Account temporarily locked E2: Account inactive → Admin notification required		
Special Requirements	Password hashing, HTTPS encryption		
Assumptions	Email delivery functional		
Frequency of Use	500 logins/day		

Use Case 3: Place Order

Use Case ID	UC006		
Use Case Name	Place Order		
Priority	10 (Critical)		
Created By	Mr. ABC	Last Updated By	January 28th, 2026

Date Created	January 19th, 2026	Last revision date	January 25th, 2026
Actor	Primary - Farmer, Secondary - Payment Gateway		
Description	Farmer completes purchase transaction with multiple payment options.		
Pre-condition	Farmer logged in, items in cart		
Post-condition	Order created, payment processed, confirmation sent		
Trigger	Farmer clicks "Checkout"		
Normal Flow of events/ Basic Flow/ Happy Path	1. Farmer reviews cart contents 2. Farmer enters/selects delivery address 3. Farmer selects payment method (COD/Card/UPI) 4. System processes payment via gateway 5. System generates Order ID and sends confirmation email 6. Order appears in "My Orders"		
Alternative Flow	A1: Apply promo code → Discount calculated A2: Save address → Added to profile		
Exceptions	E1: Payment declined → Retry option shown E2: Out of stock → Item removed or alternatives suggested		
Special Requirements	State regulatory Compliant payment gateway, COD logistics integration		
Assumptions	Payment gateway operational		
Frequency of Use	200 orders/day		

Use Case 4: Manufacturer Upload Products

Use Case ID	UC009		
Use Case Name	Manufacturer Upload Products		
Priority	9 (High)		
Created By	Mr. ABC	Last Updated By	January 28th, 2026
Date Created	January 19th, 2026	Last revision date	January 25th, 2026
Actor	Manufacturer, Secondary - Admin System		

Description	Manufacturers add fertilizers, seeds, pesticides to catalog.
Pre-condition	Manufacturer logged into dashboard
Post-condition	Product queued for approval, visible post-approval
Trigger	Manufacturer clicks "Add Product"
Normal Flow of events/ Basic Flow/ Happy Path	1. Manufacturer selects product category (fertilizer/seed/pesticide) 2. Manufacturer enters details: name, specs, price, images, stock 3. System validates mandatory fields and formats 4. Product saved as "Pending Approval" 5. Admin approves → Product live in catalog
Alternative Flow	A1: Bulk upload via CSV → Multiple products processed A2: Save draft → Continue later
Exceptions	E1: Invalid image format → Error with guidelines E2: Duplicate product → Merge suggestion
Special Requirements	Image compression, 5MB limit per image
Assumptions	Manufacturers have digital product info ready
Frequency of Use	50 uploads/week

Use Case 5: Send Order Confirmation Notifications

Use Case ID	UC008		
Use Case Name	Send Order Confirmation Notifications		
Priority	8 (Medium-High)		
Created By	Mr. ABC	Last Updated By	January 28th, 2026
Date Created	January 19th, 2026	Last revision date	January 25th, 2026
Actor	Primary - System, Secondary - Farmer/Manufacturer, Email/SMS Service		
Description	Farmers receives notification for order confirmation in real-time.		
Pre-condition	Order placed and payment processed		
Post-condition	Confirmation email and SMS delivered to customer, tracking ID generated		

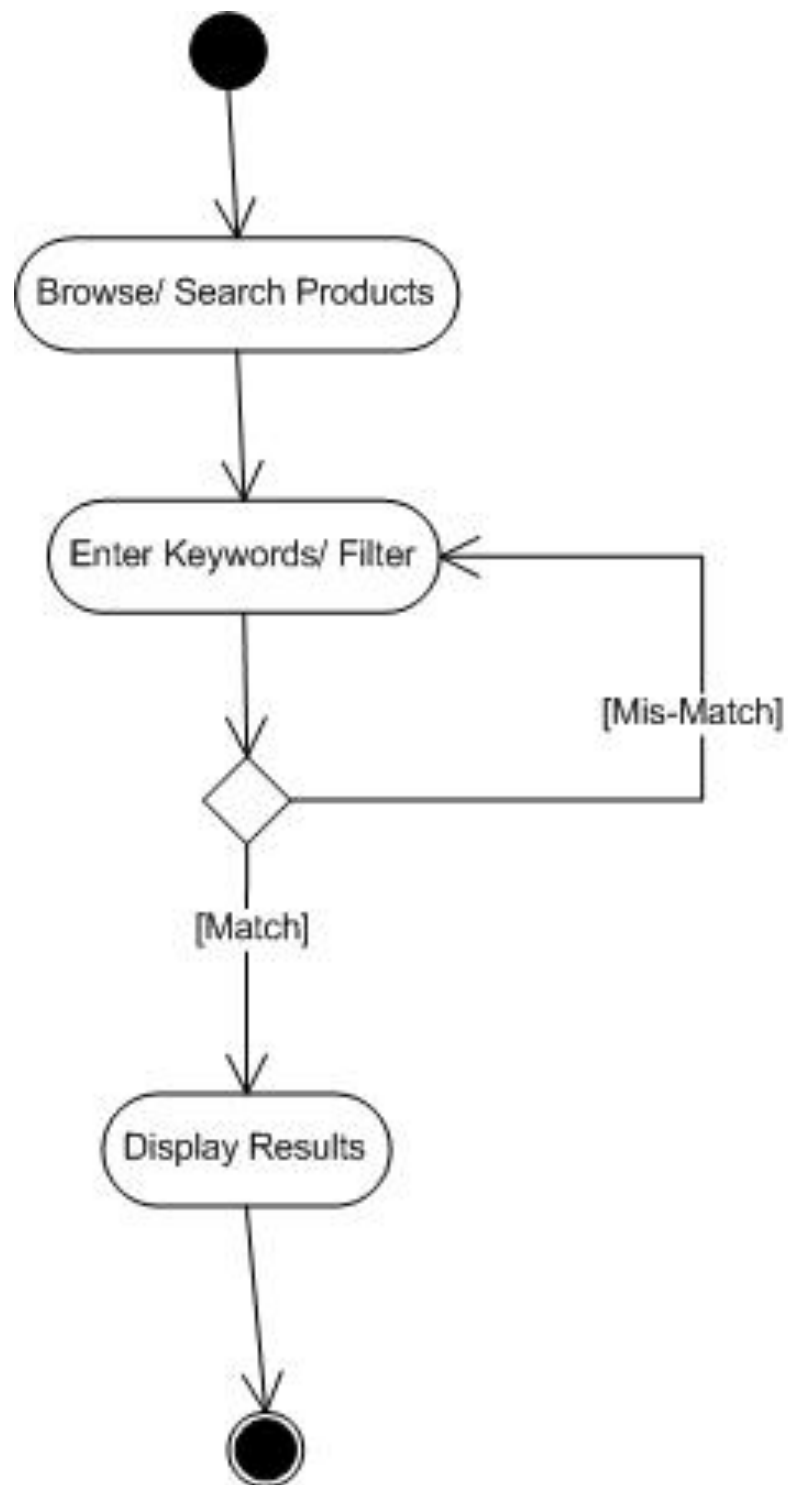
Trigger	Order status changes to "Confirmed"
Normal Flow of events/ Basic Flow/ Happy Path	1.System receives "Order Confirmed" event from payment gateway 2.System retrieves customer details (email, phone) from order database 3.System generates confirmation email with order details (Order ID, items, total, tracking link) 4.System sends email AND SMS notification in parallel 5.Email service provider processes and delivers confirmation email 6.SMS gateway delivers order summary text message 7.System logs delivery status for both notifications 8.System updates order record with confirmation timestamps
Alternative Flow	A1: Customer opted for SMS only → Skip email generation A2: Include tracking link → Hyperlink embedded in both messages A3: Bulk order → Summary email with itemized PDF attachment
Exceptions	E1: Email delivery fails → Log error, SMS still sent, notify admin E2: SMS quota exceeded → Email sent as fallback, retry SMS later E3: Both fail → Mark order "Unconfirmed", manual intervention required
Special Requirements	Real-time API integration, mobile notifications
Assumptions	Email service and SMS gateway operational
Frequency of Use	200 notifications/day

Q12. Activity diagrams

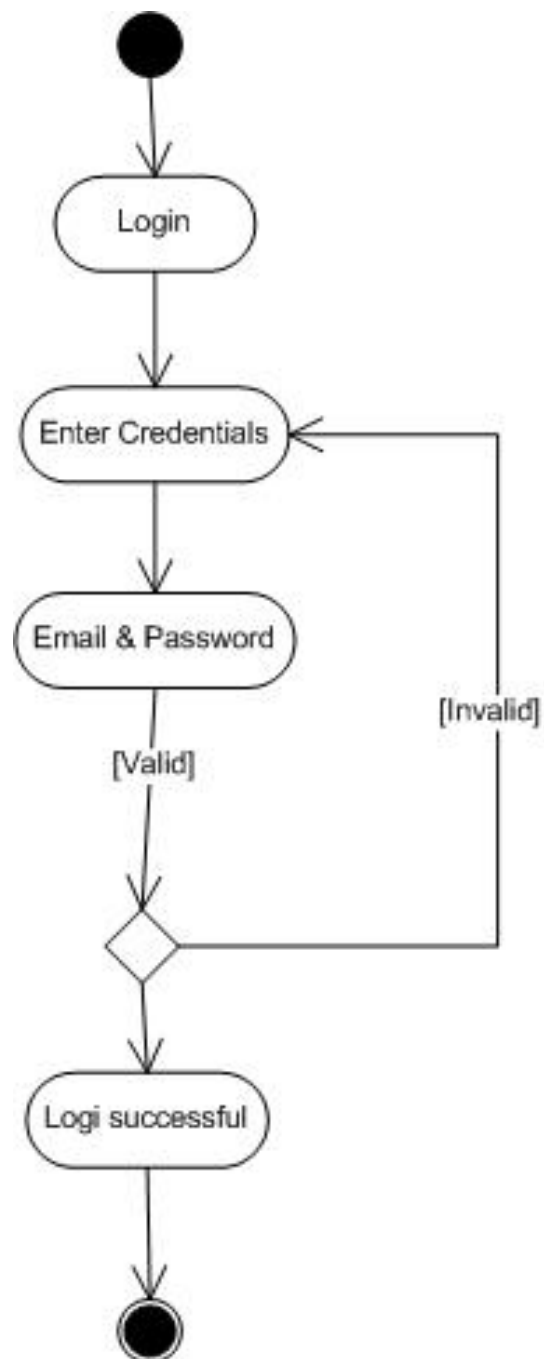
Diagrams created in MS Visio

- Components: Rounded Rectangle (Action), Diamond (Decision), Filled Circle (star node), Circle within circle (End node)
- Connectors: Control flow (arrows)
- Branch & Merge: It handles decision flows
 - Branch (Decision Node): Single input → Multiple alternate outputs based on conditions.
 - Merge Node: Multiple inputs → Single output
- Fork and Join: It handles concurrent flows. Shown by Thick horizontal/vertical bar
 - Fork Node: Single input → Multiple concurrent outputs.
 - Join Node: Multiple concurrent inputs → Single output

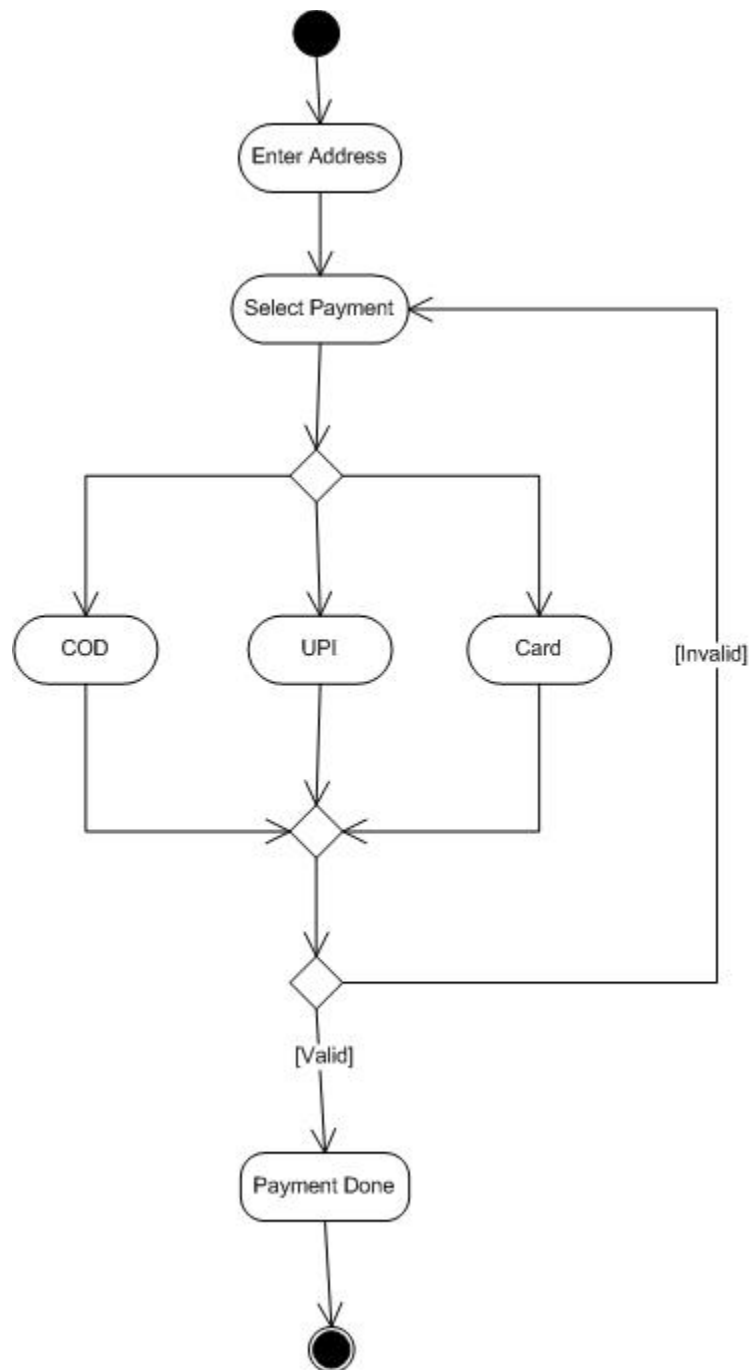
1. Farmer Search Products (BR001 - Priority 10)



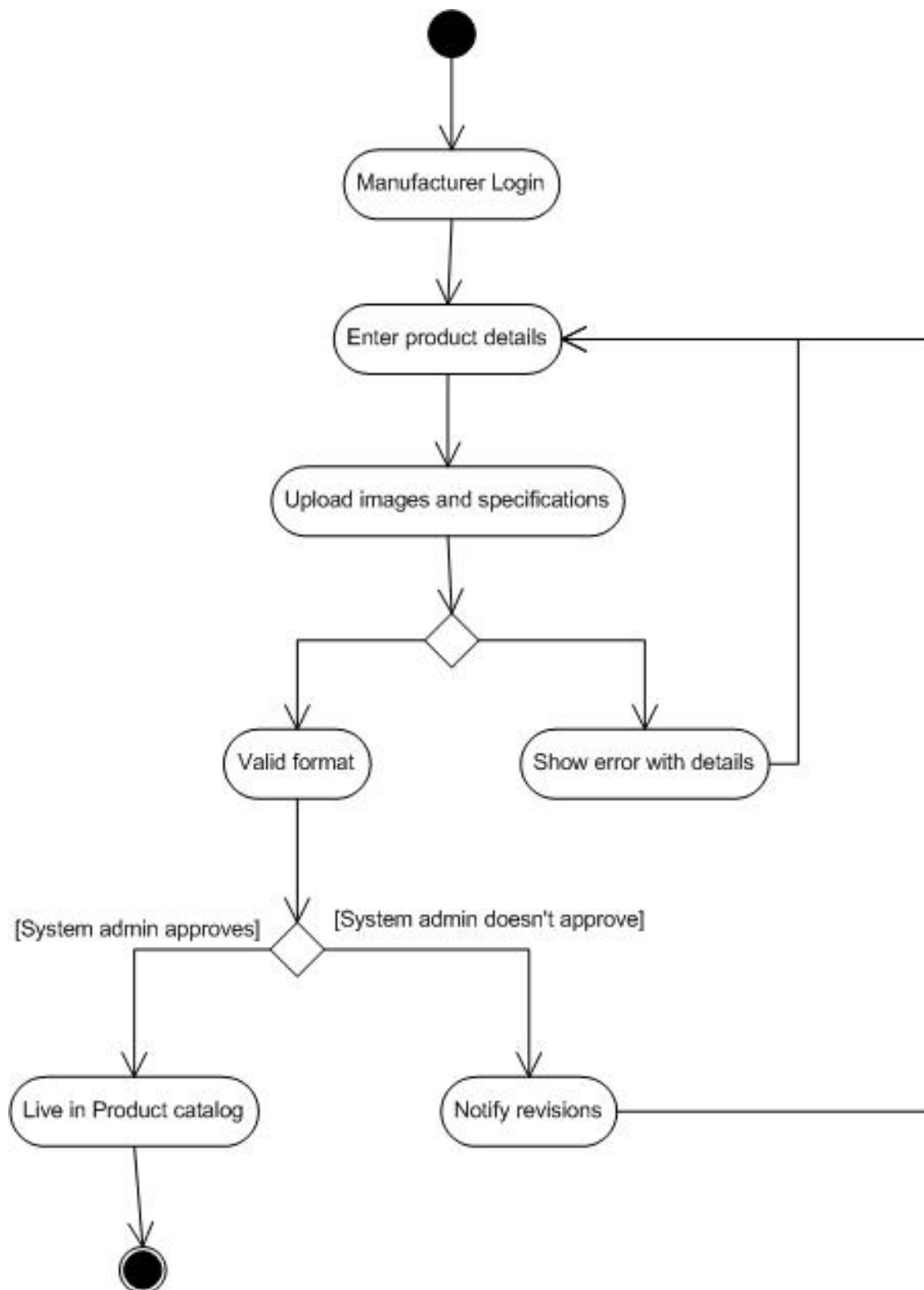
2. User Login Authentication (BR003 - Priority 10)



3. Place Order & Payment (BR007 - Priority 10)



4. Manufacturer Product Upload (BR002 - Priority 9)



5. Order confirmation email and SMS (BR008 - Priority 9)

